

HOSPITAL EMERGENCY ROOM ANALYTICS

Technical Project Report

Operational Performance Analysis — April 2023 to October 2024

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Project Type: Individual / Personal Data Analytics Project

Data Source: Publicly sourced Kaggle dataset

Primary Analytical Tool: Microsoft Excel

Project Period: April 2023 – October 2024

Deliverable: Emergency Room KPI Dashboard and Analytical Report

Professional Scope Statement

This report documents an independently developed portfolio project using a publicly available Kaggle dataset. It is not a company project, client engagement, hospital assignment, or proprietary business analysis.

1. Project Overview

The Hospital Emergency Room Analytics project is an individual Business Intelligence and data-analytics portfolio project developed to demonstrate an end-to-end workflow for transforming patient-level operational data into a structured analytical reporting solution. The project uses a publicly sourced Kaggle Emergency Room dataset and applies Microsoft Excel techniques for data preparation, aggregation, KPI development, visualization, dashboard reporting, and business interpretation.

The source dataset contains 9,216 unique patient records covering 01 April 2023 through 30 October 2024, representing 19 monthly reporting periods. The analysis evaluates patient volume, admission outcomes, waiting-time performance, patient satisfaction, demographics, and department referral patterns across the full project period.

The dashboard supplied with the project provides year/month filtering and is visually presented as a monthly operational report. The included screenshot corresponds to the December 2023 dashboard view; the broader report expands the analysis across the full April 2023–October 2024 dataset.

2. Business Problem Statement

Emergency Room operations generate high-volume patient records containing information on admissions, waiting time, demographics, referrals, and patient experience. Raw records alone make it difficult to identify demand patterns, service-level pressure points, changes over time, and data-quality issues.

The analytical problem addressed by this individual project is to convert a multi-period patient dataset into a management-oriented reporting framework capable of answering key operational questions: How does patient demand change over time? What proportion of patients are admitted? How frequently do waits exceed 30 minutes? How does performance vary by month? What demographic and referral patterns are visible? And what does the available satisfaction data indicate?

3. Business Objectives

- Analyze Emergency Room patient volume across April 2023 to October 2024.
- Compare monthly and yearly patient-volume patterns.
- Measure admission and non-admission outcomes.
- Evaluate average waiting time and identify months with elevated delays.
- Track the proportion of patients exceeding the 30-minute waiting-time threshold.
- Analyze available patient satisfaction scores and their response coverage.
- Understand patient distribution by age, gender, race, and department referral.
- Develop management-style KPIs and an Excel dashboard for operational review.
- Identify data-quality gaps and translate analytical findings into practical recommendations.

4. Dataset Overview

The project uses a publicly available Kaggle Emergency Room dataset containing 9,216 patient-level records. The data covers 01 April 2023 to 30 October 2024 and contains 9,216 unique Patient IDs with no duplicate Patient IDs identified.

Dataset Attribute	Value
Source	Kaggle — publicly sourced dataset
Total Records	9,216
Unique Patients	9,216
Project Period	01 April 2023 – 30 October 2024
Monthly Periods	19 months
Patient Gender Values	M, F, Male, Female
Department Referral	7 named departments + missing values

Satisfaction Coverage	2,517 of 9,216 records (27.3%)
Wait Time Range	10–60 minutes

4.1 Key Data Fields

Field	Analytical Purpose
Patient Id	Unique patient counting and validation.
Patient Admission Date	Monthly and time-series analysis.
Patient First Initial / Patient Last Name	Source identification fields; not used in aggregated reporting.
Patient Gender	Gender segmentation after label normalization.
Patient Age	Age-group segmentation.
Patient Race	Demographic segmentation.
Department Referral	Referral distribution and data-completeness analysis.
Patient Admission Flag	Admission outcome.
Patient Satisfaction Score	Patient-experience KPI.
Patient Waittime	Waiting-time KPI and delay classification.

5. Tools & Technologies Used

Tool / Technique	Application
Microsoft Excel	Primary platform for data preparation, calculations, reporting and dashboard presentation.
Excel Tables / Structured Data	Organized patient-level records for analysis.
Pivot-style aggregation	Monthly, demographic and referral summaries.
Excel formulas / conditional logic	KPI calculations and status classifications.
Charts / Dashboard Components	Trend, distribution and KPI visualization.
Kaggle	Public source of the underlying dataset.

The report intentionally does not claim use of SQL, Python, Power BI, or any other technology unless it was part of this project. The documented primary analytical platform is Microsoft Excel.

6. Data Preparation & Cleaning

- Validated the source record structure, field names, data types, and date coverage.
- Converted Patient Admission Date from text representation into a consistent datetime field.
- Confirmed the analytical period from April 2023 through October 2024.
- Validated Patient ID uniqueness; no duplicate Patient IDs were identified in the supplied dataset.
- Normalized gender labels from M/F and Male/Female into Male/Female reporting categories.
- Retained missing department referrals as an explicit 'No Department Referral' category rather than silently removing records.
- Excluded blank satisfaction scores from satisfaction averages rather than treating missing values as zero.
- Converted age, waiting time, and satisfaction fields to numeric analytical types.
- Created ten-year age groups from 0–9 through 70–79.
- Created a delay classification where Patient Waittime >30 minutes is Delay and ≤30 minutes is On-time.
- Created monthly reporting periods for trend analysis.

6.1 Data Quality Observations

Department Referral is missing for 5,400 records (58.6%), while Patient Satisfaction Score is missing for 6,699 records (72.7%). These missing values are retained as documented limitations rather than being treated as zero or silently discarded.

7. Analytical Methodology

The project follows a structured BI workflow:

- Raw Kaggle Dataset
- Data Validation and Cleaning
- Date and Category Transformation
- Monthly Aggregation
- KPI Calculation
- Trend and Segment Analysis
- Dashboard Visualization
- Business Interpretation
- Recommendations

7.1 KPI Definitions

KPI	Definition
Total Patients	Distinct Patient ID count for the reporting period.
Admission Rate	Admitted patients ÷ total patients.
Average Wait Time	Mean Patient Waittime.
Delay Rate	Patients with wait time >30 minutes ÷ total patients.
On-time Rate	Patients with wait time ≤30 minutes ÷ total patients.
Average Satisfaction	Mean non-blank Patient Satisfaction Score.
Satisfaction Response Coverage	Non-blank satisfaction records ÷ total records.
Age Group	Ten-year age bands from 0–9 to 70–79.

8. Detailed Analysis & Insights

8.1 Full-Period Patient Volume

Across the full April 2023–October 2024 period, the dataset contains 9,216 patient encounters. Monthly patient volume ranged from 431 in 2024-02 to 530 in 2024-08.

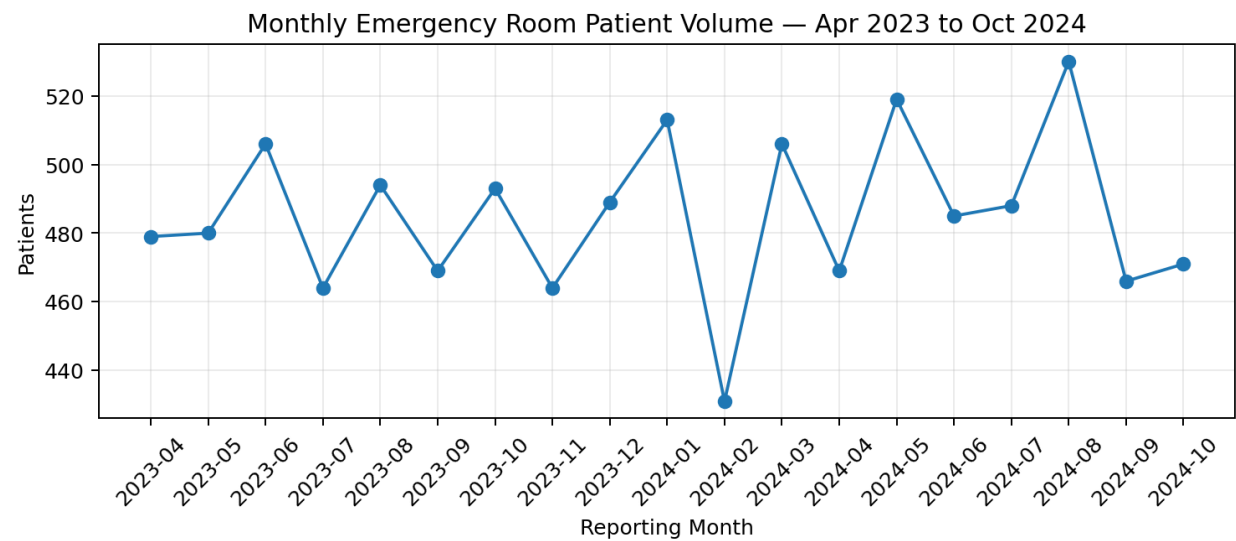


Figure 1. Monthly Emergency Room patient volume across the full project period.

8.2 Annual Performance Comparison

Year	Patients	Admissions	Admission Rate	Avg Wait	Delay Rate	Avg Satisfaction
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2023	4,338	2,157	49.72%	35.03 min	58.00%	5.03/10
2024	4,878	2,455	50.33%	35.47 min	60.50%	4.96/10

The 2024 portion of the dataset shows a slightly higher admission rate and a higher delay rate than the 2023 portion. Because the dataset covers only April–December in 2023 and January–October in 2024, the annual figures should be interpreted as reporting-period comparisons rather than complete calendar-year comparisons.

8.3 Waiting-Time Performance

Across all 9,216 encounters, average waiting time was 35.26 minutes and median waiting time was 35 minutes. Waiting time ranged from 10 to 60 minutes. The highest monthly average wait was 36.67 minutes in 2024-02, while the lowest was 34.05 minutes in 2024-10.

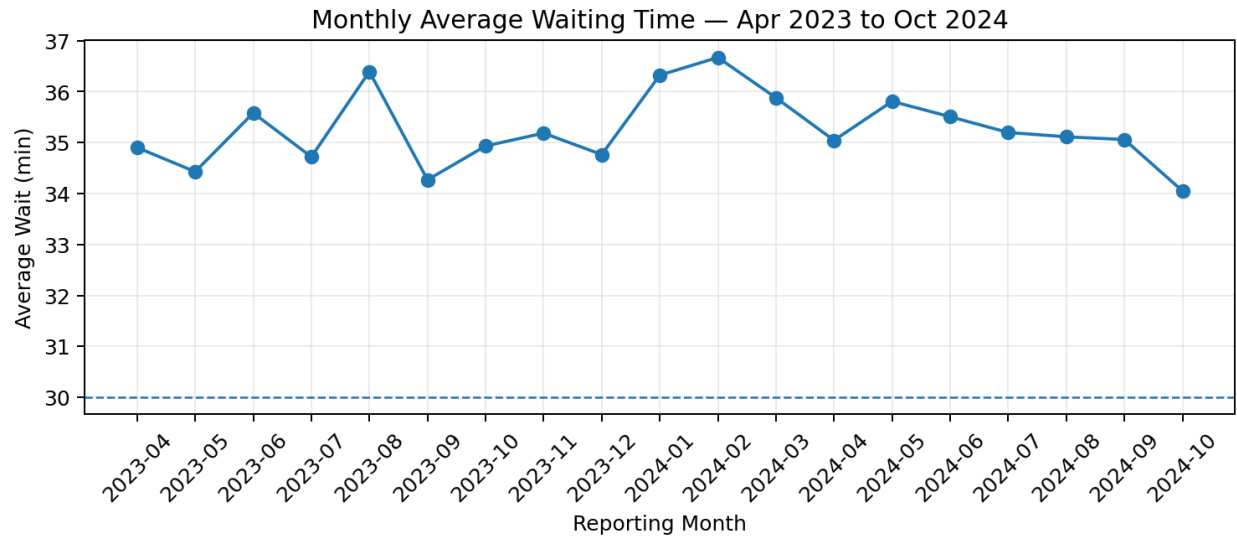


Figure 2. Monthly average waiting time; dashed line represents the 30-minute operational threshold.

8.4 Delay / On-Time Analysis

Using the project threshold of 30 minutes, 5,467 encounters (59.32%) were classified as delayed and 3,749 (40.68%) as on-time. The highest monthly delay rate was 65.66% in 2024-02, while the lowest was 54.80% in 2023-09.

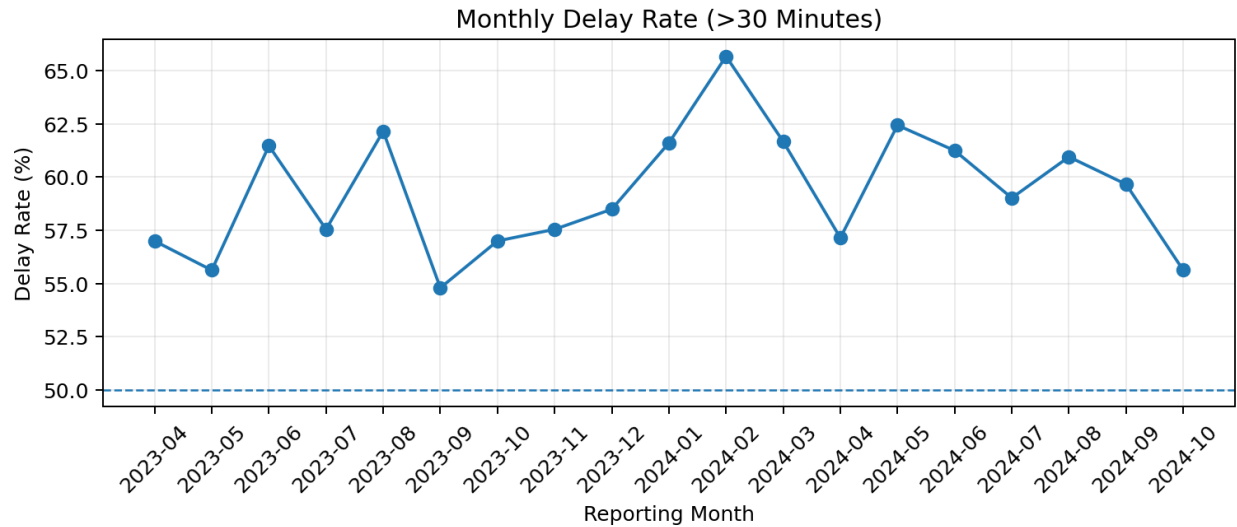


Figure 3. Monthly delay rate (>30 minutes).

8.5 Admission Analysis

Overall, 4,612 patients (50.04%) were admitted and 4,604 (49.96%) were not admitted. The overall admission split is close to 50/50, indicating that both pathways represent a substantial share of the operational workload.

8.6 Patient Satisfaction Analysis

The overall average recorded satisfaction score is 4.99/10 based on 2,517 responses, representing 27.3% of all records. Monthly average satisfaction ranged from 4.63/10 in 2024-04 to 5.33/10 in 2024-03.

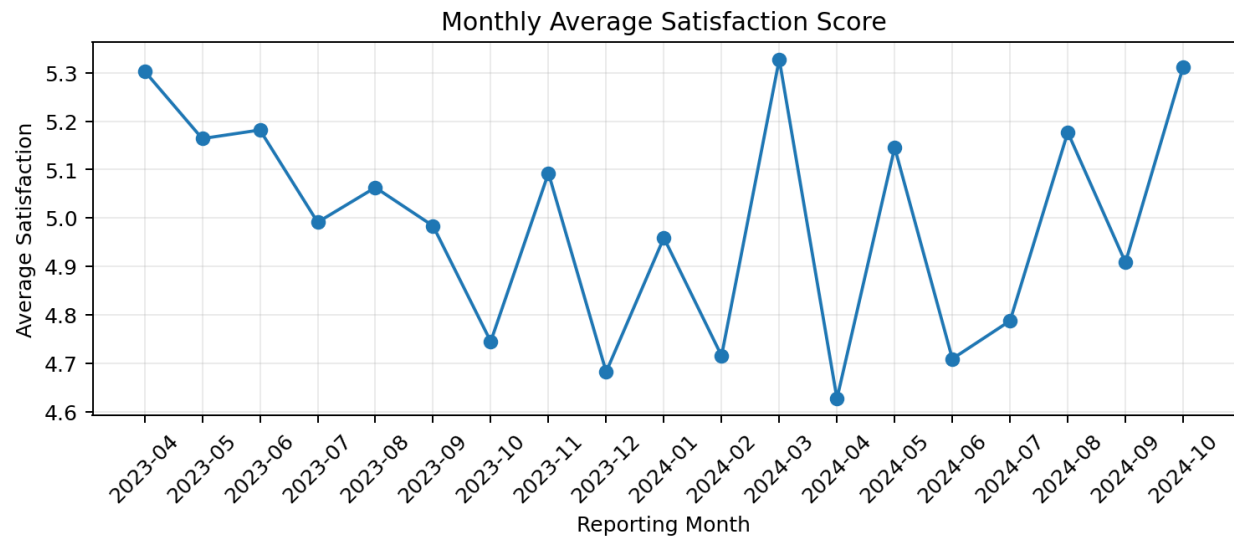


Figure 4. Monthly average satisfaction score. Blank satisfaction values are excluded.

8.7 Age Group Analysis

Age Group	Patients	Share
0-9	1056	11.5%
10-19	1179	12.8%
20-29	1188	12.9%
30-39	1200	13.0%
40-49	1135	12.3%
50-59	1151	12.5%
60-69	1154	12.5%
70-79	1153	12.5%

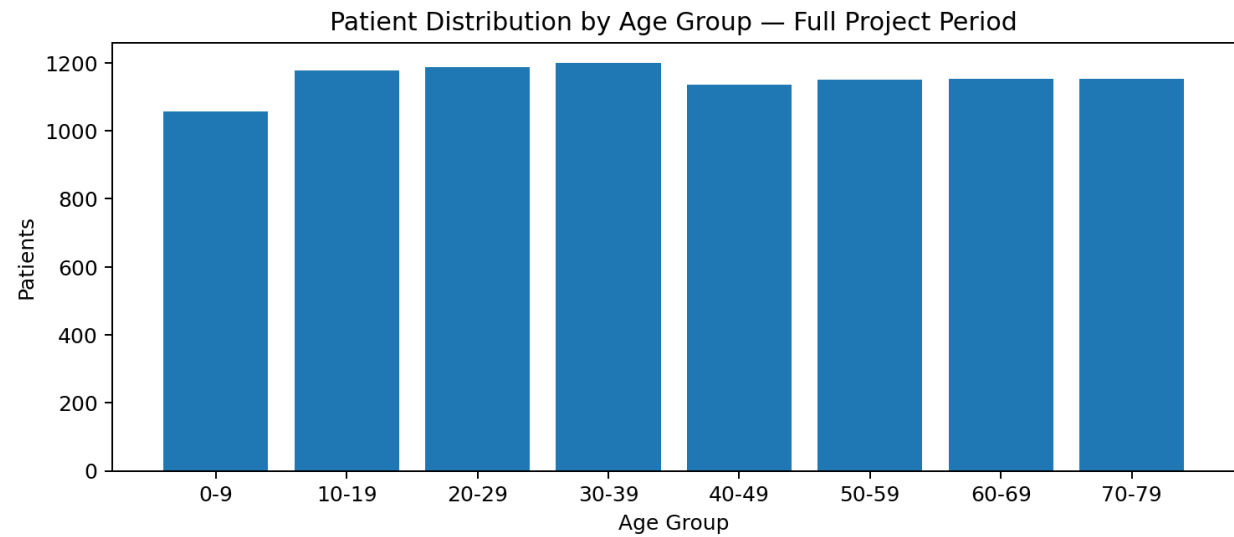


Figure 5. Patient distribution by age group across the full project period.

8.8 Gender Analysis

The normalized dataset contains 4,729 male patients (51.3%) and 4,487 female patients (48.7%). The overall distribution is broadly balanced.

8.9 Race / Demographic Analysis

Race Category	Patients	Share
White	2571	27.9%
African American	1951	21.2%
Two or More Races	1557	16.9%
Asian	1060	11.5%
Declined to Identify	1030	11.2%
Pacific Islander	549	6.0%
Native American/Alaska Native	498	5.4%

Race is included as a descriptive demographic dimension available in the source dataset. The project does not use demographic attributes to make clinical or eligibility decisions.

8.10 Department Referral Analysis

Referral Category	Patients	Share
No Department Referral	5400	58.6%
General Practice	1840	20.0%
Orthopedics	995	10.8%
Physiotherapy	276	3.0%
Cardiology	248	2.7%
Neurology	193	2.1%
Gastroenterology	178	1.9%
Renal	86	0.9%

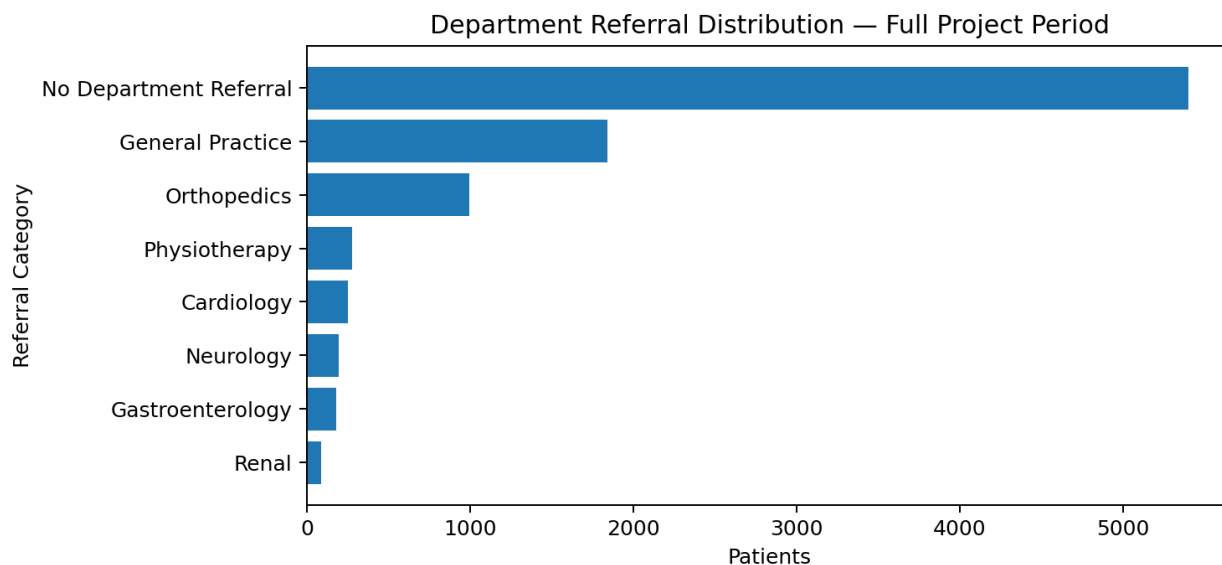


Figure 6. Department referral distribution across the full project period.

9. Key Performance Indicators (KPIs)

KPI	Full Project Period	Business Meaning
Total Patients	9,216	Overall workload across Apr 2023–Oct 2024.

Admission Rate	50.04%	Share of encounters resulting in admission.
Average Wait Time	35.26 min	Overall service-time indicator.
Median Wait Time	35 min	Central waiting-time reference.
Delay Rate	59.32%	Share exceeding 30 minutes.
On-time Rate	40.68%	Share at or below 30 minutes.
Average Satisfaction	4.99/10	Recorded patient-experience KPI.
Satisfaction Response Coverage	27.3%	Share of encounters with a recorded score.
Unique Patient IDs	9,216	Patient-level uniqueness check.

10. Dashboard Overview

The supplied Excel dashboard is designed as a monthly operational reporting interface with year and month selectors. The screenshot provided with the project has the 2023 year selector and December month selector active, and its displayed metrics match the December 2023 slice of the source dataset.

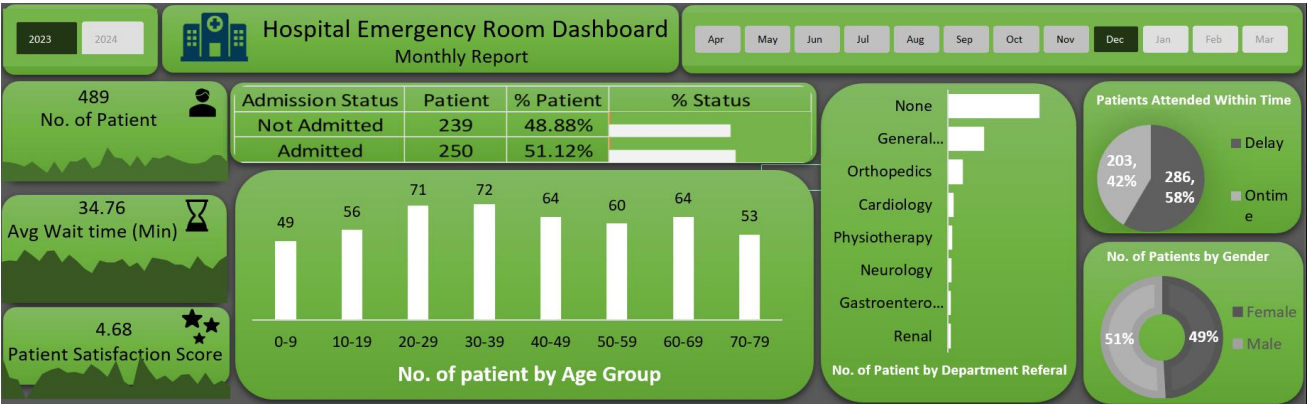


Figure 7. Supplied dashboard screenshot — December 2023 monthly view.

Dashboard Component	Purpose
Patient Volume KPI	Headline monthly workload.
Average Wait Time	Monthly service-time indicator.
Patient Satisfaction Score	Monthly experience indicator.
Admission Status	Admitted vs. not admitted.
Age Group Chart	Monthly demographic segmentation.
Department Referral Chart	Monthly referral mix, including no referral.
Patients Attended Within Time	Delay vs. on-time view using the 30-minute threshold.
Gender Chart	Monthly demographic segmentation.
Year / Month Selectors	Dashboard period navigation.

Dashboard validation: the displayed December 2023 snapshot contains 489 patients, 250 admitted, 239 not admitted, average wait 34.76 minutes, average satisfaction 4.68/10, 286 delayed, and 203 on-time. These values correspond to the December 2023 records in the supplied Kaggle dataset.

11. Business Analysis & Management Summary

- The dataset shows sustained monthly demand, with patient volume ranging from 431 to 530 encounters.
- Waiting-time performance remains a key service-level signal, with an overall delay rate of 59.32%.
- February 2024 recorded both the highest monthly average waiting time (36.67 minutes) and the highest delay rate (65.66%).
- October 2024 recorded the lowest monthly average waiting time (34.05 minutes) in the available project period.
- Admission outcomes are broadly balanced overall, with 50.04% of encounters classified as admitted.

- Department referral information is incomplete for 5,400 records (58.6%), which materially limits department-level interpretation.
- Satisfaction scores are available for only 2,517 records (27.3%), so satisfaction should be interpreted with response coverage in mind.
- The monthly dashboard format supports period-by-period review, while the full dataset enables broader trend analysis beyond a single monthly snapshot.

12. Business Recommendations

Priority Area	Recommendation	Expected Analytical Value
Waiting-Time Management	Track monthly and daily >30-minute delay rates; investigate periods with sustained elevated waits.	Improved service-level monitoring and root-cause analysis.
Peak Demand Planning	Use monthly and daily volume trends to identify high-demand periods and evaluate capacity alignment.	Better demand-to-capacity planning.
Referral Data Quality	Standardize department referral capture and apply validation rules to reduce missing values.	More reliable department-level insights.
Satisfaction Data Coverage	Improve consistency of satisfaction-score collection.	More representative patient-experience KPI.
Monthly Performance Review	Use a standardized monthly KPI pack covering volume, admission, wait time, delay, and satisfaction.	Consistent management reporting.
Exception Reporting	Flag months/days with high volume, high delay rate, or low satisfaction.	Faster identification of operational pressure points.
Dashboard Drill-down	Add filters for date, age group, department, admission status, and delay status.	Faster segment-level investigation.

13. Data Quality, Limitations & Governance

- The source contains 9,216 unique patient records covering April 2023 to October 2024.
- The annual comparison is partial-year: 2023 contains April–December and 2024 contains January–October, so it should not be treated as a full-calendar-year comparison.
- Department Referral is missing for 5,400 records (58.6%).
- Patient Satisfaction Score is missing for 6,699 records (72.7%); satisfaction averages use non-blank scores only.
- Gender values contain abbreviated and full labels and were normalized for reporting.
- The source includes patient-name fields; they are not used in the aggregated report to minimize unnecessary exposure of identifiable information.
- The analysis is descriptive and operational. It is not a clinical decision-support system and should not be interpreted as patient-care guidance.
- The dataset is publicly sourced from Kaggle. No proprietary, confidential, or employer-owned hospital data is claimed.
- All findings are intended for portfolio demonstration, learning, and analytical practice.

14. Conclusion

This individual Hospital Emergency Room Analytics project demonstrates an end-to-end data analytics workflow across a multi-period dataset spanning April 2023 to October 2024. The analysis covers 9,216 patient records and transforms raw operational data into structured KPIs, monthly trends, demographic analysis, referral analysis, dashboard views, and business recommendations.

Across the full period, the average waiting time was 35.26 minutes and 59.32% of encounters exceeded the 30-minute threshold. Monthly analysis highlights February 2024 as a key pressure period for waiting time and delay rate, while October 2024 shows comparatively lower average waiting time.

The project also demonstrates practical data-analytics discipline through data validation, missing-value treatment, category normalization, time-series analysis, KPI definition, and dashboard interpretation. The supplied December 2023 dashboard snapshot is used as a monthly example within the broader April 2023–October 2024 analysis rather than being mistaken for the full project period.

As a personal portfolio project, the deliverable demonstrates practical capability in Microsoft Excel analytics, data preparation, KPI development, dashboard reporting, trend analysis, data-quality assessment, and business-oriented storytelling.

15. Appendix — Resume / Portfolio Project Description

Hospital Emergency Room Analytics Dashboard | Microsoft Excel | Kaggle Dataset | Apr 2023 – Oct 2024

Developed an individual Excel-based Emergency Room analytics dashboard using a publicly sourced Kaggle dataset containing 9,216 patient records across April 2023–October 2024. Analyzed patient volume, admission outcomes, average wait time, >30-minute delay rate, satisfaction, age, gender, race, and department referrals. Built monthly KPI summaries and dashboard visualizations, performed data cleaning and validation, analyzed multi-period operational trends, and translated findings into recommendations focused on waiting-time management, capacity planning, referral-data quality, satisfaction coverage, and exception reporting.